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Studierichting: Informatica

Oefeningenreeks 5B nr. 3.2

$$\begin{aligned} & \int_0^1 \frac{\ln x}{(x+1)^2} dx \\ &= \lim_{b \rightarrow 0+} \int_b^1 \ln x (x+1)^{-2} dx \end{aligned}$$

We weten dat: $\int_d^c u dv = [uv]_d^c - \int_d^c v du$

Laat:

$$u = \ln x, \quad dv = (x+1)^{-2} dx$$

$$du = \frac{1}{x} dx, \quad v = -(x+1)^{-1}$$

Dus:

$$\lim_{b \rightarrow 0+} \int_b^1 \ln x (x+1)^{-2} dx = \lim_{b \rightarrow 0+} \left(- \left[\frac{\ln x}{x+1} \right]_b^1 + \int_b^1 \frac{1}{x(x+1)} dx \right)$$

We weten dat:

$$\frac{1}{x(x+1)} = \frac{A}{x} + \frac{B}{x+1}$$

$$\Rightarrow A = 1, \quad B = -1$$

$$\begin{aligned} \lim_{b \rightarrow 0+} \int_b^1 \ln x (x+1)^{-2} dx &= \lim_{b \rightarrow 0+} \left(- \left[\frac{\ln x}{x+1} \right]_b^1 + \int_b^1 \left(\frac{1}{x} - \frac{1}{x+1} \right) dx \right) \\ &= \lim_{b \rightarrow 0+} \left(- \left(\frac{\ln 1}{2} - \frac{\ln b}{b+1} \right) + [\ln x - \ln(x+1)]_b^1 \right) \\ &= \lim_{b \rightarrow 0+} \frac{\ln b}{b+1} + \ln 1 - \ln 2 - \ln b + \ln(b+1) \\ &= \lim_{b \rightarrow 0+} \frac{\ln b}{b+1} - \ln b + \ln(b+1) - \ln 2 \\ &= \lim_{b \rightarrow 0+} \ln b \left(\frac{1}{b+1} - 1 \right) + \ln(b+1) - \ln 2 \\ &= \lim_{b \rightarrow 0+} \ln b \left(\frac{-b}{b+1} \right) + \ln(b+1) - \ln 2 \\ &= 0 + 0 - \ln 2 \\ &= -\ln 2 \end{aligned}$$

References